

Module 7 — Practice Task

Introduction to JavaScript

Task 1 — Your First JavaScript Program

Create a file named app.js and print three introductory messages.

```
console.log("Hello JavaScript!");  
console.log("My name is Alex.");  
console.log("I am excited to become a Web Developer.");
```

Expected Console Output

```
Hello JavaScript!  
My name is Alex.  
I am excited to become a Web Developer.
```

Task 2 — Simple Math Calculator

Create two number variables and print the results of the five basic arithmetic operations.

```
let num1 = 25;  
let num2 = 10;  
  
console.log("Addition:", num1 + num2);  
console.log("Subtraction:", num1 - num2);  
console.log("Multiplication:", num1 * num2);  
console.log("Division:", num1 / num2);  
console.log("Modulus:", num1 % num2);
```

Expected Console Output

```
Addition: 35  
Subtraction: 15  
Multiplication: 250  
Division: 2.5  
Modulus: 5
```

Task 3 — Student Information

Store student details in variables of the appropriate type and print them.

```
let studentName = "Alex Johnson";  
let age = 20;  
let country = "Bangladesh";  
let isStudent = true;
```

```
console.log("Student Name:", studentName);
console.log("Age:", age);
console.log("Country:", country);
console.log("Is Student:", isStudent);
```

Expected Console Output

```
Student Name: Alex Johnson
Age: 20
Country: Bangladesh
Is Student: true
```

Task 4 — Identify Data Types

Use the `typeof` operator to check the data type of each variable.

```
let a = 100;
let b = "Programming";
let c = true;
let d = false;
let e = 3.1416;

console.log("a:", typeof a);
console.log("b:", typeof b);
console.log("c:", typeof c);
console.log("d:", typeof d);
console.log("e:", typeof e);
```

Expected Console Output

```
a: number
b: string
c: boolean
d: boolean
e: number
```

Task 5 — Variable Naming Challenge

Only names that follow JavaScript's identifier rules (start with a letter, \$, or `_`; no hyphens; not a reserved keyword) are valid.

Valid names:

- `firstName` — valid (camelCase)
- `First_Name` — valid (letters and underscore allowed)
- `user123` — valid (numbers allowed after the first character)
- `123user` — invalid (cannot start with a number)
- `my-name` — invalid (hyphens are not allowed in identifiers)
- `$price` — valid (\$ is a legal starting character)
- `_total` — valid (_ is a legal starting character)

- `let` — invalid (reserved keyword)

Code using only the valid names:

```
let firstName = "Alex";
let First_Name = "Alex";
let user123 = 1;
let $price = 19.99;
let _total = 0;

console.log(firstName, First_Name, user123, $price, _total);
```

Task 6 — Personal Bio

Store personal details and print a friendly, formatted introduction.

```
let name = "Alex";
let age = 20;
let favoriteFood = "Pizza";
let favoriteLanguage = "JavaScript";
let lovesCoding = true;

console.log("Hello!");
console.log("My name is " + name + ".");
console.log("I am " + age + " years old.");
console.log("My favorite food is " + favoriteFood + ".");
console.log("My favorite programming language is " + favoriteLanguage + ".");
console.log("I love coding:", lovesCoding);
```

Expected Console Output

```
Hello!
My name is Alex.
I am 20 years old.
My favorite food is Pizza.
My favorite programming language is JavaScript.
I love coding: true
```

Task 7 — Arithmetic Practice

Take two numbers and print the results of all five arithmetic operations, matching the example format.

```
let number1 = 50;
let number2 = 8;

console.log("Number 1:", number1);
console.log("Number 2:", number2);
console.log();
console.log("Addition:", number1 + number2);
console.log("Subtraction:", number1 - number2);
console.log("Multiplication:", number1 * number2);
console.log("Division:", number1 / number2);
```

```
console.log("Modulus:", number1 % number2);
```

Expected Console Output

```
Number 1: 50  
Number 2: 8  
  
Addition: 58  
Subtraction: 42  
Multiplication: 400  
Division: 6.25  
Modulus: 2
```

Task 8 — Shorthand Operators

Apply each shorthand operator to score and print the value after every step.

```
let score = 50;  
  
score += 10;  
console.log("After += 10:", score);  
  
score -= 5;  
console.log("After -= 5:", score);  
  
score *= 2;  
console.log("After *= 2:", score);  
  
score /= 5;  
console.log("After /= 5:", score);  
  
score %= 4;  
console.log("After %= 4:", score);
```

Expected Console Output

```
After += 10: 60  
After -= 5: 55  
After *= 2: 110  
After /= 5: 22  
After %= 4: 2
```

Note: the exact results depend on the numbers you choose to apply each operator with — any consistent sequence is correct.

Task 9 — Mini Coding Challenge

Calculate the total price, discount amount, and final price for a purchase.

```
let price = 200;  
let quantity = 3;  
let discount = 10; // percent
```

```
let totalPrice = price * quantity;
let discountAmount = (totalPrice * discount) / 100;
let finalPrice = totalPrice - discountAmount;

console.log("Total Price:", totalPrice);
console.log("Discount Amount:", discountAmount);
console.log("Final Price:", finalPrice);
```

Expected Console Output

```
Total Price: 600
Discount Amount: 60
Final Price: 540
```

Task 10 — Console Quiz (Answers)

Question 1

```
let x = 20; let y = 5; console.log(x + y);
```

Answer: 25

Question 2

```
let language = "JavaScript"; console.log(typeof language);
```

Answer: string

Question 3

```
let isOnline = false; console.log(typeof isOnline);
```

Answer: boolean

Question 4

```
let num = 10; num += 5; console.log(num);
```

Answer: 15

Question 5

```
let a = 30; let b = 7; console.log(a % b);
```

Answer: 2

Mini Project — Personal Introduction Card

A complete program that stores personal information and prints a formatted profile card.

```
let fullName = "John Doe";
let age = 22;
let country = "Bangladesh";
```

```

let favoriteLanguage = "JavaScript";
let dreamJob = "Full-Stack Developer";
let yearsToGoal = 1;
let isLearningJS = true;

console.log("=====");
console.log("      MY PROFILE");
console.log("=====");
console.log();
console.log("Name:", fullName);
console.log("Age:", age);
console.log("Country:", country);
console.log("Favorite Language:", favoriteLanguage);
console.log("Dream Job:", dreamJob);
console.log("Years to Reach Goal:", yearsToGoal);
console.log("Learning JavaScript:", isLearningJS);
console.log();
console.log("=====");

```

Expected Console Output

```

=====
      MY PROFILE
=====

Name: John Doe
Age: 22
Country: Bangladesh
Favorite Language: JavaScript
Dream Job: Full-Stack Developer
Years to Reach Goal: 1
Learning JavaScript: true

=====

```

Bonus Challenge — Shopping Bill Calculator

Calculate a subtotal, tax amount, and grand total, then print a neat bill.

```

let productName = "Wireless Mouse";
let price = 15;
let quantity = 4;
let taxPercent = 5;

let subtotal = price * quantity;
let taxAmount = (subtotal * taxPercent) / 100;
let grandTotal = subtotal + taxAmount;

console.log("---- Shopping Bill ----");
console.log("Product:", productName);
console.log("Price per unit:", price);
console.log("Quantity:", quantity);
console.log("Subtotal:", subtotal);
console.log("Tax (" + taxPercent + "%):", taxAmount);
console.log("Grand Total:", grandTotal);

```

```
console.log("-----");
```

Expected Console Output

```
---- Shopping Bill ----  
Product: Wireless Mouse  
Price per unit: 15  
Quantity: 4  
Subtotal: 60  
Tax (5%): 3  
Grand Total: 63  
-----
```

Learning Checklist

- Create and run a JavaScript file
- Declare and use variables
- Choose appropriate variable names
- Work with Number, String, and Boolean data types
- Use typeof to identify data types
- Perform arithmetic operations
- Use shorthand operators (+=, -=, *=, /=, %=)
- Build a small JavaScript program using variables and calculations